

What to take to the desktop machine

Castelnuovo di Porto nonlinear time-history · target machine: Intel i9-9900KF, 8 cores, 32 GB RAM.

Full instructions: [docs/Castelnuovo_Run_On_Desktop.pdf](#), which is inside the repository once it is cloned.

1. Carry on a USB stick — one file only

☐	File	Size	Where it is now
☐	<code>castelnuovo-image.tar.gz</code> the Docker image: OpenSees 3.8.0 + MUMPS, gms, the whole Python environment	1.40 GB 1 496 262 271 bytes	<code>C:\Users\mlaur\Desktop\</code>

Verify the copy before trusting it. A 1.4 GB USB transfer can truncate silently. On the desktop, in PowerShell, in the folder where you copied it:

```
Get-FileHash castelnuovo-image.tar.gz -Algorithm SHA256
```

The hash must be exactly:

```
FF893903453EBC636B3CD8C8B62D29FF8B5E44B2CE6F3F1286F8EEC52D468089
```

PowerShell prints it in uppercase; the comparison is case-insensitive. If it differs, copy the file again — do not proceed.

2. Nothing else needs carrying — it all comes from GitHub

☐	What	Included in the clone
☐	Analysis code, geometry, material data	yes
☐	Seismic record — Montenegro 1979, Run 2.1 longitudinal, PGA 0.593 g <code>resources/records/castelnuovo/montenegro1979_run21_long.txt</code> — already in m/s ² , do not rescale	yes
☐	Reference values for the verification step <code>resources/reference/castelnuovo_tied_10modes.json</code>	yes
☐	The step-by-step guide (PDF) <code>docs/Castelnuovo_Run_On_Desktop.pdf</code>	yes

```
git clone https://github.com/mlaura1996/modularBIMtoFEM.git
cd modularBIMtoFEM
git checkout chapter7-castelnuovo
```

The branch is `chapter7-castelnuovo`, not `main`. Last commit as pushed: `67dfac8`.

3. To install on the desktop (needs internet, ~15 min)

☐	Software	Notes
☐	Docker Desktop	Accept the WSL 2 backend; reboot; start it and wait for the whale icon to stop animating
☐	Git for Windows	All defaults
☐	File <code>C:\Users\<user>\.wslconfig</code>	Contents below. Then <code>wsl --shutdown</code> and restart Docker Desktop. Skipping this is the usual cause of the analysis being killed with no error message.

```
[wsl2]
memory=24GB
```

processors=8
swap=8GB

4. Free disk space needed on the desktop

Item	Space
Docker image once loaded (it decompresses)	~7 GB
Repository clone	~1 GB
Analysis output (recorders, sampled)	allow 20–30 GB
Total, with margin	~50 GB

5. Order of operations — do not reorder

1. Install Docker Desktop + Git, write `.wslconfig`, restart.
2. `docker load -i <path>\castelnuovo-image.tar.gz`
3. Clone the repository and check out `chapter7-castelnuovo`.
4. **Run the verification** — `verify_environment.py`, ~20 min. It must print `PASS`. Do not skip it.
5. **Run the smoke test** — `timehistory_castelnuovo.py --smoke`, ~10 min. Two lines matter; see below.
6. Only then, the full run, detached.

6. Two things to report back before the full run starts

(a) The conflict guard. The smoke test prints either `conflict guard passed` or `ABORT: N node(s) are both tied and duplicated at a contact interface`. If it aborts, stop and report it — it is an anticipated modelling conflict, not something to work around.

(b) The recorder check. Find the block `--- recorder check after 5 steps ---` and send back the list of any recorder marked `EMPTY`. The damage-output response names for this material could not be verified in advance; OpenSees writes a file holding only the time column instead of raising an error, so a 30-hour run could otherwise finish with no damage data at all.

Expected run time: roughly 15–40 hours for the 19.5 s window (3900 steps). The first progress line of the transient gives a real estimate for that machine. The setup stages before it — meshing, then 20–60 minutes building one damage material per element — are normal and not a hang.

The 8 cores will not speed this up: the solver is sequential. And **there is no restart** — if the run is interrupted it begins again from zero, so disable Windows automatic restarts and sleep first.

After copying the image to the desktop, delete `castelnuovo-image.tar.gz` from the laptop Desktop — that drive is at 97% capacity.